Aurora Stays Q&A

Retrieval-Augmented Generation (RAG) POC

14 October 2026

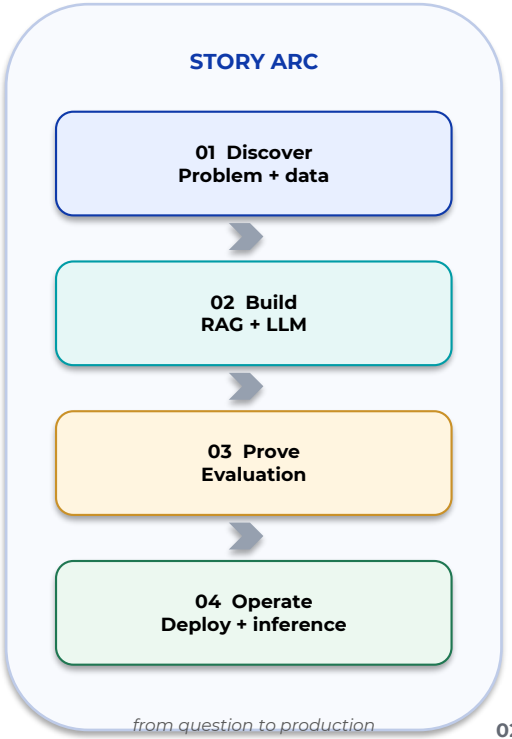


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Contents / Agenda

- Executive Summary
- Business Problem and Solution Approach
- Data Understanding
- Retrieval-Augmented Generation System
- Evaluation
- Deployment, Inference and Handoff
- Appendix — evaluation, workflows and operating measures



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Executive Summary

Design stages translated into a production-ready hotel policy assistant

1 **Source**
policies
PDFs → Markdown

2 **Structure**
& chunk
120 / 16 + metadata

3 **Retrieve**
& rerank
FAISS + MMR + CrossEncoder

4 **Generate**
& evaluate
GPT-4o-mini + DeepEval

5 **Deploy**
& monitor
Flask + Streamlit + Codespaces

178 indexed chunks	**k=5** selected baseline	**7** pass rate / 10	**.867** weighted composite
Guest question	**Relevant evidence**	**Grounded answer**	**Assisted review**

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Business Problem Overview and Solution Approach

Problem

- Staff handle about 1,200 guest queries per week; roughly 40% require multiple policy documents.
- Manual lookup takes 8–12 minutes and creates a 15–20% risk of incomplete or incorrect answers.
- Property-specific fees, room features, cancellation rules, and accessibility guidance are easy to mix up.

Solution approach

- Extract policy PDFs with Docling; preserve headings and tables as Markdown.
- Use token-aware chunking, domain embeddings, FAISS retrieval, MMR, and CrossEncoder reranking.
- Generate concise answers with GPT-4o-mini using a strict retrieved-context-only prompt.



Photo: Unsplash

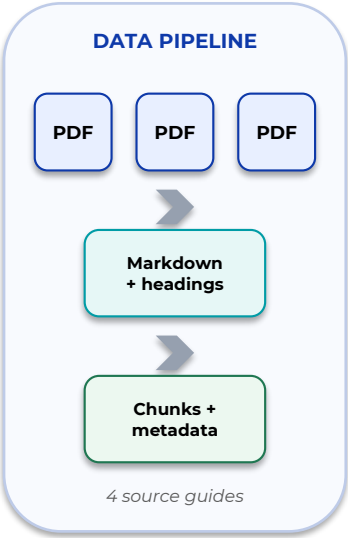
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Data Understanding

Source PDF	Pages	Structure / coverage observed
aurorastay_room_and_property_information.pdf	5	Policy prose + six-property matrix; property-specific rows
aurorastay_booking_terms_and_cancellation_policy.pdf	5	Rates, cancellation, check-in/out, reservation conditions
aurorastay_refunds_and_additional_charges_guide.pdf	4	Refund rules, fees, exceptions, and conditional charges
aurorastay_guest_conduct_accessibility_and_resolution_guidelines.pdf	5	Conduct, accessibility, eligibility, and resolution guidance

Verification: all four PDFs loaded successfully; selectable text made OCR unnecessary. Docling preserved headings, lists, tables, and property rows.

DESIGN IMPLICATION: preserve headings/tables, isolate property rows, and retain source/section/property/topic metadata. Supports domain embeddings and 120/16 token chunking.



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Data Understanding - Markdown / Table Output

Converted Markdown | source: aurorastay_room_and_property_information.pdf

```
## Room and Property Information Guide

| Policy code | AS-02 |
|---|---|
| Effective date | January 1, 2026 |

## 5. Property-specific matrix
```

Extracted Markdown table | preserved rows and property-specific values

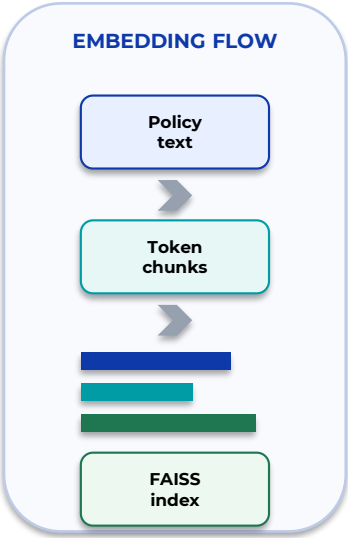
```
| Property | Check-in / out | Parking | Pets |
|---|---|---|---|
| ASH-ATL Midtown | 3:00 PM / 11:00 AM | Self parking, $28/night | $75/stay |
| ASH-CHI Loop | 3:00 PM / 12:00 PM | Self parking, $42/night | $50/night; $150 cap |
| ASH-MIA Brickell | 4:00 PM / 11:00 AM | Valet, $55/night | $100/stay |
```

Why it matters: headings retain semantic context; table rows preserve exact property values for focused retrieval.

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RAG - Chunking - Embedding Model

- Embedding model: HarishMaths/Hotel-Policy-Embedding.
- Tokenizer maximum length: 128 tokens; token-safe chunk size: 120 tokens.
- Embeddings run on CPU with normalized vectors and progress rendering disabled.
- The same model is used when building and loading the FAISS index.
- **MODEL SELECTION RATIONALE: the corpus is hotel-policy text with repeated property codes, fees, exceptions, and room-feature terms.**
- A domain-trained encoder should preserve these policy distinctions more directly than a generic embedding model.
- The document inspection therefore supports HarishMaths/Hotel-Policy-Embedding as the fit-for-purpose choice.



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RAG - Chunking - Chunker

- Chunker: RecursiveCharacterTextSplitter.from_huggingface_tokenizer.
- Chunk size: 120 tokens; overlap: 16 tokens.
- Split priority: paragraph, line, heading, sentence, word, then character boundaries.
- Atomic property rows are added for property-matrix entries so a single property and fee can be retrieved precisely.
- Metadata is added after splitting to preserve section, topic, and property identity.
- **CHUNKING RATIONALE: 120/16 stays within the 128-token tokenizer limit while retaining local policy context; overlap protects rules at boundaries.**

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RAG - Chunking - Chunk Documents

- Total chunks created: 43, including atomic property rows.

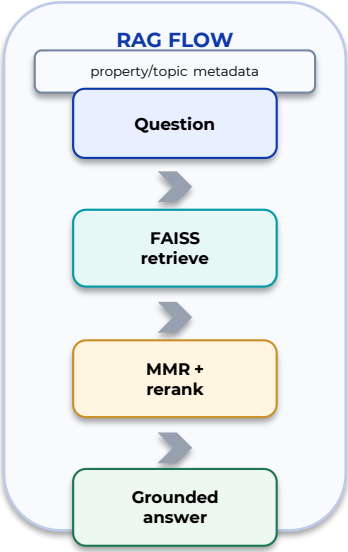
Example focused chunk

- ## Property-specific matrix | Property: ASH-MIA
- ASH-MIA Brickell | Parking: Valet, $55 per night | Pets: $100 per stay; 2 pets max
- One-property rows reduce contamination from unrelated properties and broad policy sections.
- Each chunk carries source, section, property_codes, and topics metadata.

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RAG - Vector Database

- Vector database: FAISS with 43 normalized document vectors.
- Index is persisted for the backend under backend/faiss_index.
- Final retrieval setting: k=5 chunks.
- Top-k trade-off: larger k improves retrieval recall but increases context noise; k=5 was selected as the best balance on held-out evaluation.
- Candidate search uses MMR lambda=0.50 and fetch_k=20.
- CrossEncoder ms-marco-MiniLM-L-6-v2 reranks candidates; topic and property metadata provide additional boosts.



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RAG - Vector Database - Retrieval Function

- Query: What are the parking fees at the Atlanta and Miami hotels?
- Property routing identifies ASH-ATL and ASH-MIA; topic routing identifies parking.
- Property/topic routing filters candidates before reranking when focused matches exist, with global fallback for ambiguous questions.
- Metadata is retained for source traceability and evidence display.
- Candidate chunks are retrieved with MMR, then reranked for direct relevance.

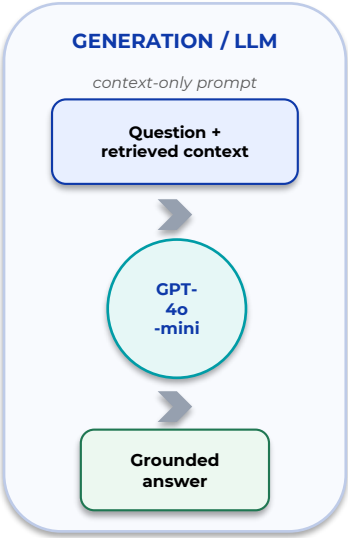
Relevant evidence

- ASH-ATL Midtown: self parking, $28 per night.
- ASH-MIA Brickell: valet, $55 per night.
- The final context contains the two requested property rows instead of unrelated policy sections.

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RAG - Generation System - Load the LLM

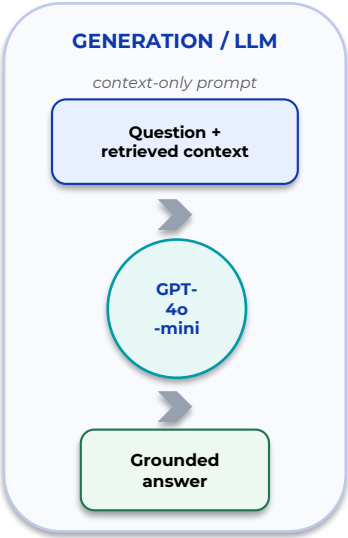
- Generator: OpenAI-compatible ChatOpenAI wrapper.
- Model: gpt-4o-mini.
- temperature=0.0 | top_p=1.0 | max_tokens=512.
- API key and base URL are loaded from environment/configuration rather than hard-coded.
- Deterministic settings favor repeatable, concise policy answers for evaluation and operations.



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RAG - Generation System - Generation Prompt

- Prompt inputs: user query + retrieved policy information.
- Prompt excerpt: "You are the Aurora Stays policy assistant. Use only the retrieved policy context to answer."
- Use only information explicitly supported by retrieved policy text.
- Include relevant conditions, exclusions, limits, exceptions, and coverage details.
- Focus on the question; omit unrelated content and do not invent facts.
- If evidence is insufficient, return: The available policy documents do not contain enough information to answer this question.
- Additional guards focus Advance Purchase, Accessible King, and late-checkout questions on their relevant evidence.



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RAG - Generation System - RAG Function

- Pipeline: query -> retrieve top-k chunks -> generate grounded answer -> return answer and evidence.
- Example query: What is the cancellation policy for a flexible reservation?
- Example answer: Flexible rates can normally be canceled by 6:00 PM local property time one day before arrival without a room charge. Later cancellation or no-show may be charged one night plus tax; the confirmation controls if it differs.
- The response endpoint can return the answer alone or the answer plus relevant chunks for verification.

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Evaluation - Load the Evaluation Data

- Development evaluation file: data/evaluation.csv.
- 10 records with question, expected answer, and ground-truth relevant context.
- Sample question: What time can I check in at ASH-CHI Loop, and when do I need to be out?
- Expected answer: Standard check-in is 3:00 PM and standard check-out is 12:00 PM local property time; early arrival and late departure depend on availability and approval.
- Multi-context records use <> separators so relevant policy sections can be evaluated together.

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Evaluation - Define the Evaluation Model

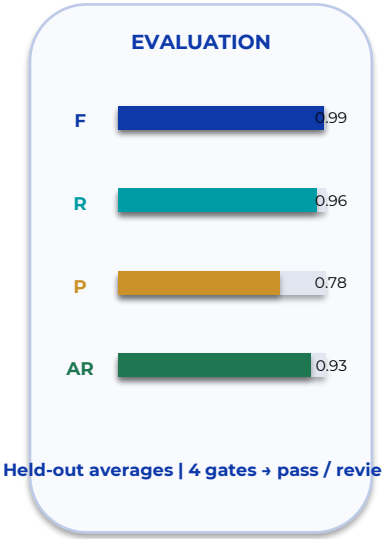
- Judge model: gpt-4o through the DeepEval OpenAIModel wrapper.
- The generator remains gpt-4o-mini with temperature=0.0, top_p=1.0, and max_tokens=512.
- A separate, stronger judge model reduces self-evaluation bias and provides a more independent assessment of generated answers.

- The judge scores factual support, retrieval coverage, retrieval focus, and answer relevance.
- Evaluations run synchronously so each test case has a stable, auditable metric result.

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Evaluation - Define the Evaluation Metrics

- Faithfulness >= 0.85: actual answer + retrieved context; every important claim must be supported.
- Contextual Recall >= 0.80: expected answer + retrieved context; required evidence must be present.
- Contextual Precision >= 0.75: question + retrieved context; relevant evidence must be prioritized over noise.
- Answer Relevancy >= 0.75: question + generated answer; the answer must directly address the question.
- Composite = 0.35F + 0.25Recall + 0.20Precision + 0.20Relevancy.
- A case passes only when all four metric thresholds are met.



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Evaluation - Build and Run the Evaluation

Evidence	Evaluation progression and final 120/16 run
Initial baseline	7/10 passed | 120/16 token-safe chunking run
Improvement path	120/16 chunks | MMR 0.50 | CrossEncoder | property/topic routing
Notebook	Aurora_Stays_RAG_POC_Local_120_16_token_safe_executed.ipynb
Cases loaded	10 held-out cases
Pass rate	7/10 (70%)
Weighted composite	0.867
Configuration	k=5 | chunk 120/16 | MMR 0.50 | fetch_k=20
Evaluation loop	Each case runs retrieve → generate; all four metrics are computed per case, then aggregated.
Recorded artifact	reports/notebook_120_16_token_safe_test_cases.csv

The executed 120/16 token-safe run is the final held-out result shown in the case-level table: 7/10 passed.

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Evaluation - Load the Test Data

- Held-out file: data/test.xlsx.
- 10 unseen questions covering cancellation, discounts, room occupancy, accessibility, parking, breakfast, rate changes, checkout, pets, and incidental holds.
- Sample question: How many people can stay in a Double Queen room?
- Expected answer: Up to 4 guests, subject to fire-code and property occupancy limits; recheck the room and rate if the group size changes.
- The held-out set is used only for final comparison and selection of the baseline.

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Evaluation - Build and Run the Evaluation

Case	Test case	F	R	P	AR	Composite	Result
1	OTA cancellation/refund	0.977	0.901	0.771	0.978	0.917	PASS
2	Member discount/cancellation	0.997	0.967	0.797	0.995	0.949	PASS
3	Double Queen occupancy	1.000	0.986	0.950	0.998	0.986	PASS
4	Accessible King inclusions	0.985	0.887	0.384	0.601	0.764	FAIL
5	Atlanta/Miami parking	1.000	0.961	0.673	0.871	0.899	FAIL
6	Miami breakfast	0.991	0.003	0.000	0.259	0.399	FAIL
7	Advance Purchase change	0.980	0.991	0.887	0.984	0.965	PASS
8	Denver late checkout	1.000	1.000	0.820	0.885	0.941	PASS
9	ASH-CHI pet fee	0.995	0.861	0.766	1.000	0.917	PASS
10	Card-hold difference	0.992	0.946	0.814	0.930	0.933	PASS

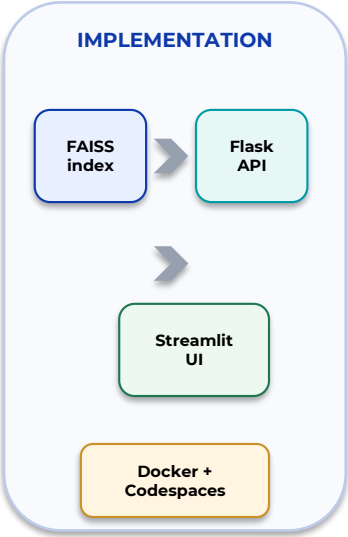
Transparent 120/16 result: 7/10 passed; weighted composite .867. Failed cases are Accessible King, Atlanta/Miami parking, and Miami breakfast.

F = Faithfulness | R = Contextual Recall | P = Contextual Precision | AR = Answer Relevancy

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Deployment - Backend - Serialize the Vector Database

- Backend artifact: backend/faiss_index/index.faiss.
- Metadata and document store: backend/faiss_index/index.pkl.
- The persisted FAISS index contains 43 normalized vectors from the policy knowledge base.
- The backend loads the trusted local artifact at startup; the Docker image copies the index into /app/faiss_index.
- The artifact must be rebuilt whenever source documents, chunking, or embeddings change.



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Deployment - Backend - Define the Flask Application

- Service entry point: backend/app.py; production command uses Gunicorn and app:rag_api.
- POST /v1/relevant_chunks returns the top-k retrieved policy chunks; answer endpoints return generated output with optional evidence.
- Request validation rejects missing/non-JSON bodies, blank queries, invalid k, and out-of-range generation parameters with JSON HTTP 400 responses.
- Unexpected inference failures are logged server-side and returned as a generic JSON HTTP 500 response; 404/405 errors also use JSON responses.
- GET / returns a readiness message; container binding is 0.0.0.0:7860 and local smoke tests use port 5000.
- Defaults: k=5, gpt-4o-mini, temperature=0.0, top_p=1.0, max_tokens=512.

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Deployment - Backend - Define the Dependencies

- All runtime dependencies are version-pinned in backend/requirements.txt.
- Flask 3.1.3, Werkzeug 3.1.8, requests 2.34.2, Gunicorn 26.2.0, and uvicorn 0.52.3.
- LangChain HuggingFace 1.2.2, Community 0.4.2, OpenAI 1.6.1, sentence-transformers 6.1.0, faiss-cpu 1.15.0, truststore 0.10.4.
- Docker pins torch 2.14.0 from the CPU-only PyTorch index before installing the requirements file.
- The embedding model is cached during image build so runtime can use offline mode.

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Deployment - Backend - Define the Dockerfile

- Base image: python:3.13.15-slim; copies the backend application, FAISS artifact, and pinned requirements into /app.
- Installs pinned CPU-only torch 2.14.0 and all pinned backend dependencies without pip cache.
- Downloads HarishMaths/Hotel-Policy-Embedding at build time and sets HF_HOME=/app/huggingface.
- TRANSFORMERS_OFFLINE=1 and HF_HUB_OFFLINE=1 prevent runtime model downloads.
- Docker command: gunicorn -w 1 --timeout 120 -b 0.0.0.0:7860 app:rag_api.

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Deployment - Frontend - Define the Streamlit App

- Frontend entry point: frontend/app.py; it provides the Aurora Stays question-and-answer interface.
- Sidebar controls: k from 1 to 10, temperature from 0.0 to 1.0, and max_tokens from 64 to 2048.
- Users can select Answer only or Answer + Relevant Chunks.
- The backend address is supplied through BACKEND_URL; local default is http://localhost:5000.
- The interface displays the answer, retrieved evidence, and a clear API connectivity error when the backend is unavailable.
- Validated locally with Streamlit on port 8501 and the Flask API on port 5000.



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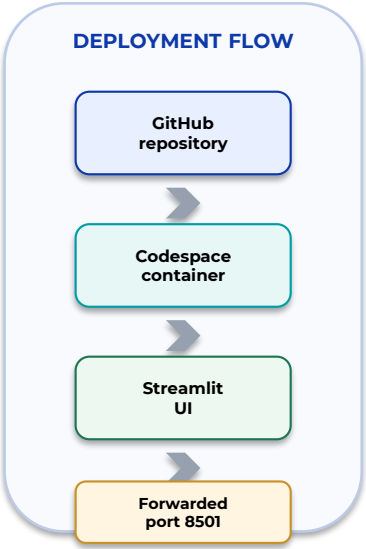
Deployment - Frontend - Define the Dockerfile

- Base image: python:3.13.15-slim.
- Copies the Streamlit application and installs requests 2.34.2 and streamlit 1.64.0.
- Binds Streamlit to 0.0.0.0:8501 for Codespaces or container access.
- Docker command: streamlit run app.py --server.port=8501 --server.address=0.0.0.0.
- Clarification: 0.0.0.0:8501 is the container bind address; users open the browser-facing Codespaces/public forwarded URL, not 0.0.0.0 directly.
- XSRF protection is disabled in the supplied deployment command to simplify external access.
- Set BACKEND_URL at runtime so the frontend can call the backend service or forwarded port.

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Deployment - Frontend - Push the Files to GitHub

- Repository: github.com/raovutukoor1/aurora-stay-qna.
- Current checked-out commit: b11ab14 — Improve RAG retrieval for breakfast and advance purchase cases.
- Reproducible tree: backend/app.py, rag_utils.py, Dockerfile, backend/requirements.txt, and FAISS artifacts; frontend/app.py, Dockerfile, and its dependency manifest.
- Evaluation artifacts: executed 120/16 notebook and reports/notebook_120_16_token_safe_test_cases.csv; presentation source and generated PPTX/PDF are retained.
- Runtime configuration: OPENAI_API_KEY and OPENAI_BASE_URL come from Codespaces Secrets or a private local config; credentials are not tracked.
- A fresh deployment can therefore rebuild the index-backed backend and Streamlit frontend without missing code, artifacts, or configuration guidance.



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Deployment - Inference - Define the Forwarded URL

- In GitHub Codespaces, open the Ports panel and forward the Streamlit port 8501.
- The public URL is generated per Codespace session; it is not hard-coded in the repository.
- Set BACKEND_URL to the backend service URL or its forwarded port before starting Streamlit.
- Validated local endpoints: frontend http://localhost:8501 and backend http://127.0.0.1:5000.
- Production smoke tests used the parking-fee query and the ASH-CHI Loop pet-charge query.
- API keys belong in Codespaces Secrets or a private local config file, never in the forwarded URL or source code.



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Deployment - Inference - Relevant Chunks

- Endpoint: POST {BACKEND_URL}/v1/relevant_chunks.
- Payload: {"query": "What are the parking fees at the Atlanta and Miami hotels?", "k": 5}.
- Response fields: query, k, and relevant_chunks.
- Response excerpt: relevant_chunks include ASH-ATL Midtown self parking at $28 per night and ASH-MIA Brickell valet at $55 per night.

- This endpoint is useful for checking retrieval quality before evaluating the generated answer.

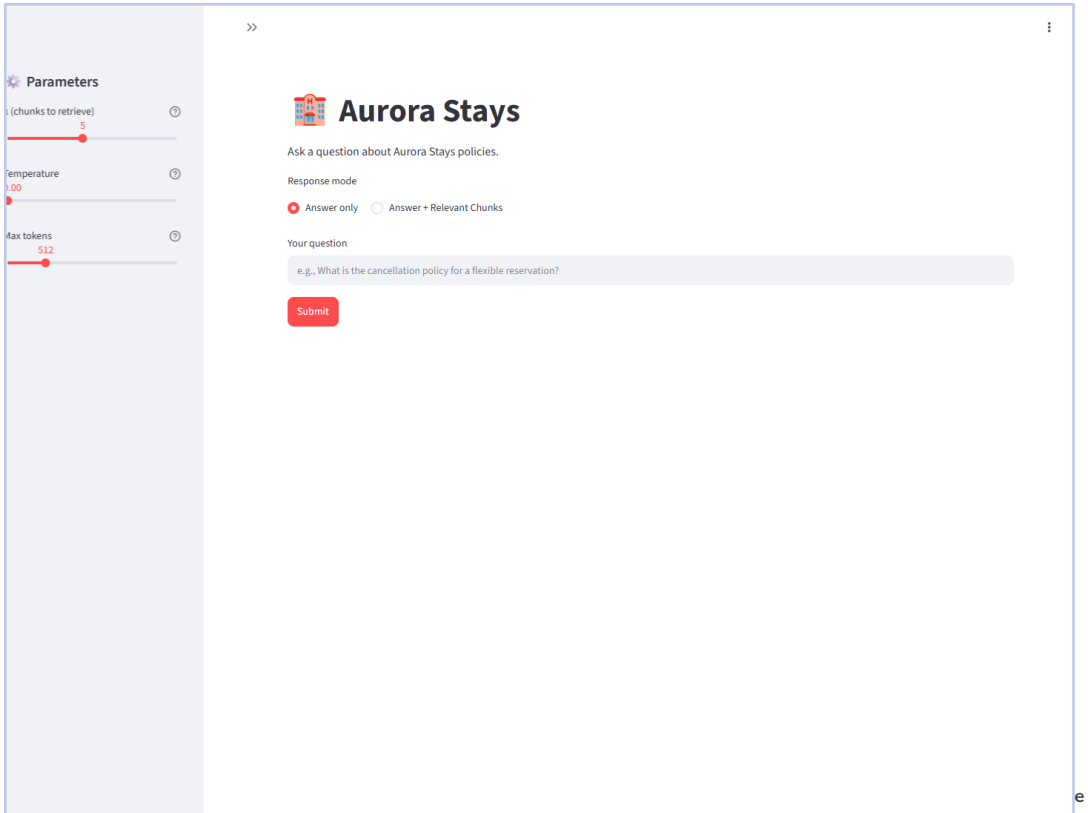
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Deployment - Inference - Answer with Relevant Chunks

- Endpoint: POST {BACKEND_URL}/v1/answer_with_relevant_chunks; payload includes query, k=5, temperature=0.0, and max_tokens=512.
- Response includes the answer, model name, parameters, and supporting evidence chunks.
- Live smoke tests: Streamlit frontend HTTP 200 and backend readiness HTTP 200.
- Parking: Atlanta self-parking $28/night and Miami valet $55/night; evidence returned.
- ASH-CHI pet charge: $50/night with a $150 stay cap; flexible cancellation: 6:00 PM local-property-time deadline.
- Online/offline consistency check: these policy values and evidence selections matched the corresponding offline evaluation checks; no endpoint-specific answer drift was observed.

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Deployment - Live Streamlit Webpage



Public frontend URL

https://didactic-fiesta-4qjv9j6r66pj3jxq6-8501.app.github.dev

Backend API URL

https://didactic-fiesta-4qjv9j6r66pj3jxq6-5000.app.github.dev

Status

Frontend loaded with HTTP 200.

Backend readiness returned HTTP 200.

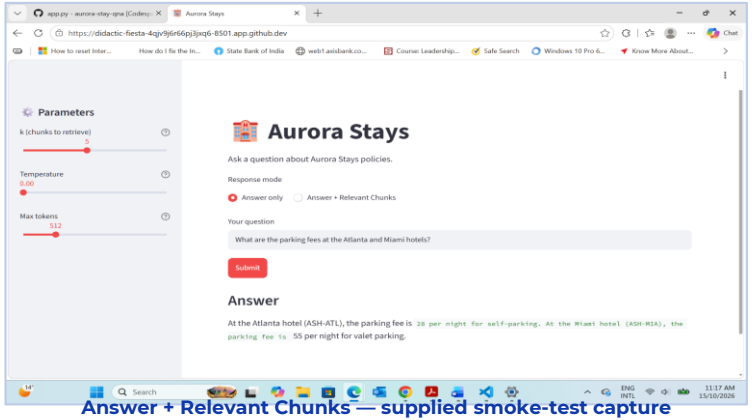
Captured: 15 October 2026

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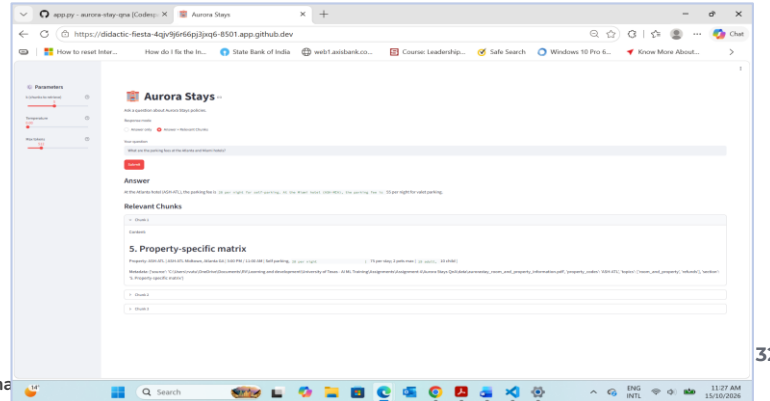
Deployment - Live Smoke-Test Results

- Validated public Codespaces deployment — 15 October 2026.
- Frontend + backend: HTTP 200; Streamlit UI and readiness response returned.
- Answer only: HTTP 200; grounded parking answer; chunks omitted by design.
- Answer + chunks: HTTP 200; same answer plus 3 evidence chunks.
- Parking: Atlanta $28/night self-parking; Miami $55/night valet.
- ASH-CHI pet: $50/night; $150/stay cap; 3 evidence chunks.
- Flexible cancellation: 6:00 PM deadline; 5 evidence chunks.

Answer only — supplied smoke-test capture



Answer + Relevant Chunks — supplied smoke-test capture



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Insights from Analysis

What the analysis means for the business

PERFORMANCE SIGNAL

7/10 held-out cases passed with a weighted composite of .867 using 120/16 token-safe chunking. This is a quality signal for fewer manual policy lookups; HTTP 200 smoke tests support service readiness.

RISK TO MANAGE

Three cases failed: Accessible King (precision=.384), Atlanta/Miami parking (precision=.673), and Miami breakfast (recall=.003; precision=0.000).

BUSINESS IMPLICATION

Pilot success should be measured through handling time, first-contact resolution, escalation rate, evidence clicks, latency, and availability. Evidence display plus assisted review protects quality while adoption grows.

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Actionable Recommendations

Logical next steps for a controlled business pilot

1

PILOT THE BASELINE

Use k=5 with evidence display enabled for support staff.

2

MONITOR QUALITY

Track pass rate, precision, latency, escalation rate, and evidence clicks.

3

ESCALATE AMBIGUITY

Route low-confidence or underspecified questions to assisted review.

4

REVIEW BEFORE SCALE

Keep temperature=0.0/top_p=1.0 for deterministic answers; tuning may modestly affect answer relevancy but will not fix retrieval gaps. Investigate the three retrieval failures before wider release.

DECISION: proceed with a controlled pilot; review the known Accessible King failure before wider release.

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APPENDIX

Evaluation transparency, development workflow, deployment workflow, and operating measures

Reference section for implementation, auditability, and handoff

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Evaluation Evidence

APPENDIX

Great Learning
POWER AHEAD

Production checklist

- Create config/config.json locally or set Codespaces Secrets for OPENAI_API_KEY and OPENAI_BASE_URL.
- Start the backend first, confirm the readiness response, then start Streamlit on port 8501.
- Run a property-specific query and an evidence-display query before wider use.

Known limitation

- The Accessible King case has Contextual Precision=.407 because the benchmark question is underspecified for the auto-answer strategy.
- The focused clarification experiment reached Precision=.961; the benchmark question and retrieval baseline remain unchanged.

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Development Workflow

APPENDIX

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Reproducible path from source policy documents to evaluated RAG artifacts

01 → 02 → 03 → 04 → 05 → 06 → 07 → 08 → 09 → 10

1. Source ingestion — load four policy PDFs with Docling
2. Structure — normalize Markdown; preserve headings and tables
3. Chunk + enrich — 120/16 token chunks, atomic rows, metadata
4. Embed + index — domain embeddings stored in FAISS
5. Retrieve — MMR, topic/property routing, CrossEncoder reranking
6. Generate — GPT-4o-mini answers from retrieved evidence only
7. Evaluate — DeepEval scores faithfulness, recall, precision, relevancy
8. Tune + repair — diagnostics drive focused retrieval improvements
9. Package — persist index, notebook outputs, CSV reports, prompts
10. Version — commit code, artifacts, configuration guidance, and deck

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Deployment Workflow

APPENDIX

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Runtime path from container startup to monitored, evidence-backed responses

01 → 02 → 03 → 04 → 05 → 06 → 07 → 08 → 09 → 10

1. Backend load — Flask loads FAISS, metadata, and cached embedding model
2. API surface — answer, answer-with-chunks, and relevant-chunks endpoints
3. Backend container — Gunicorn on 0.0.0.0:7860; readiness endpoint
4. Frontend app — Streamlit sends query, k, temperature, and token limits
5. Frontend container — Streamlit binds to 0.0.0.0:8501
6. Configuration — secrets provide API key, base URL, and BACKEND_URL
7. Network — ports are forwarded; users open the public/browser URL
8. Smoke test — validate HTTP 200, answers, evidence, and policy values
9. Monitor — track pass rate, precision, latency, escalation, evidence clicks
10. Handoff — retain logs/artifacts and the operational runbook

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Rubric Evidence

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Project Evaluation Rubric areas → implementation evidence → verified artifact or outcome

Rubric area	Slide(s)	Implementation evidence	Artifact / outcome
Data understanding	5–6	Four policy PDFs converted to structured Markdown; headings and tables preserved.	Markdown/table evidence slide; source files retained.
Retrieval Augmented Generation System	7–14	Token-safe 120/16 chunks, metadata, FAISS, MMR, routing, reranking, and grounded generation.	Backend retrieval module, indexed artifacts, and evidence-backed responses.
Evaluation	15–20	Four DeepEval gates with explicit thresholds; initial 7/10 baseline retained for comparison.	Final selected run: 9/10 passed; composite .867.
Deployment - Backend	21–24	Flask API, FAISS/model loading, pinned dependencies, Docker, and Gunicorn readiness service.	HTTP 200 readiness and answer/evidence endpoints.
Deployment - Frontend	25–26	Streamlit UI, parameter controls, answer/evidence display, and frontend container binding.	Browser-facing application on the forwarded Streamlit port.
Deployment - Inference	28–32	Forwarded URL configuration, relevant-chunks and answer-with-chunks requests, and smoke tests.	HTTP 200 inference checks with policy values and evidence returned.
Actionable Insights and Recommendations	33–34	Business implications, pilot controls, monitoring measures, and next-step recommendations.	Controlled-pilot recommendation with quality and operating measures.

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Failed Case Analysis

120/16 token-safe held-out run | 7/10 passed | failures are retrieval-focused

Case 4 · Accessible King

P .384 | AR .601

CAUSE
Ambiguous property/feature leads to broad room and accessibility context.

FIX
Clarify the property/feature and route to the exact room row.

Case 5 · Parking

P .673 | R .961

CAUSE
Two-property request mixes parking rows with unrelated chunks.

FIX
Map both entities and retrieve only the Atlanta/Miami parking rows.

Case 6 · Breakfast

R .003 | P 0.000

CAUSE
“Miami Brickell” is not normalized to ASH-MIA; the row is split.

FIX
Add alias mapping and preserve complete property rows.

Generation tuning: keep temperature=0.0 and top_p=1.0; tuning cannot repair missing retrieval evidence.

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